Digestion and Absorption (General Anatomy and Physiology of Fish)

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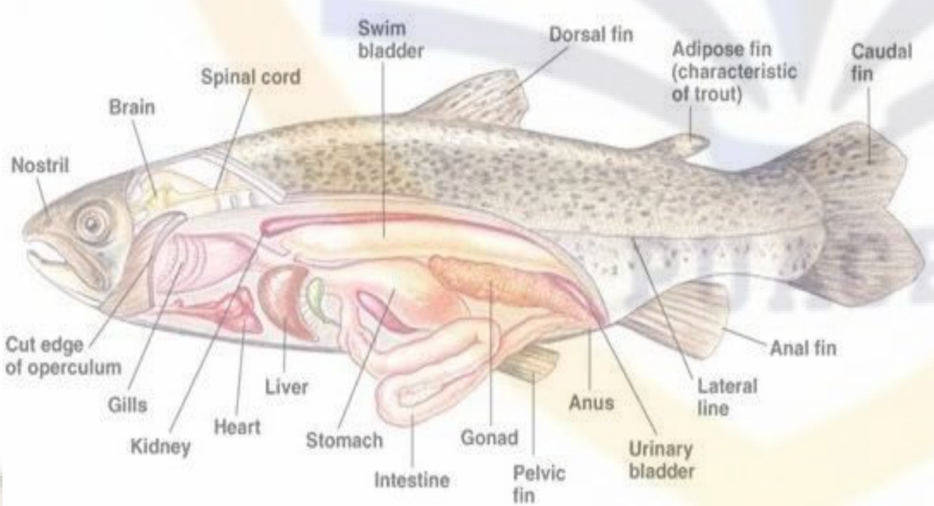
Introduction

Digestive system includes alimentary canal and associated glands.Alimentary canal or gut shows four distinct regions.

1. **Ingressive zone**
2. **Progressive zone**
3. **Degressive zone**
4. **Egressive zone**



MOUTH

OESOPHAGUS

STOMACH

INTESTINE

RECTUM

❖Mouth

Mouth is the anterior opening of alimentary canal .

Most fish mouth fall into three general types:

1 . Superior mouth type

2. Terminal mouth type

3. Inferior or sub terminal mouth type


Terminal mouth
subterminal mouth
Inferior mouth
Superior mouth

❖Teeth

- Teeth are generally hollow cones of dentine containing the pulp cavities.

- Absence of teeth in case of cyprinoids ,which have got pharyngeal teeth instead of normal teeth which are adapted for crushing & grinding the prey .

- Pharyngeal teeth are located below & above pharyngeal arches.


Teeth adapted for crushing hard shell of mollusks & crustaceans


CUTTING TEETH OF PIRANHA


CUTTING INCISORS AND GRINDING MOLARS OF PORGY


GRASPING PHARYNGEAL TEETH OF MINOW

BUCCAL CAVITY, PHARYNX & GIL RACKERS:

•The buccal cavity and pharynx are not clearly marked off from each other.

•A number of perforations of gill slits are located on each side of the pharyngeal wall.

•Primary function of the gill rakers is to protect gill filaments from injury and to assist the fishes in the process of ingestion .

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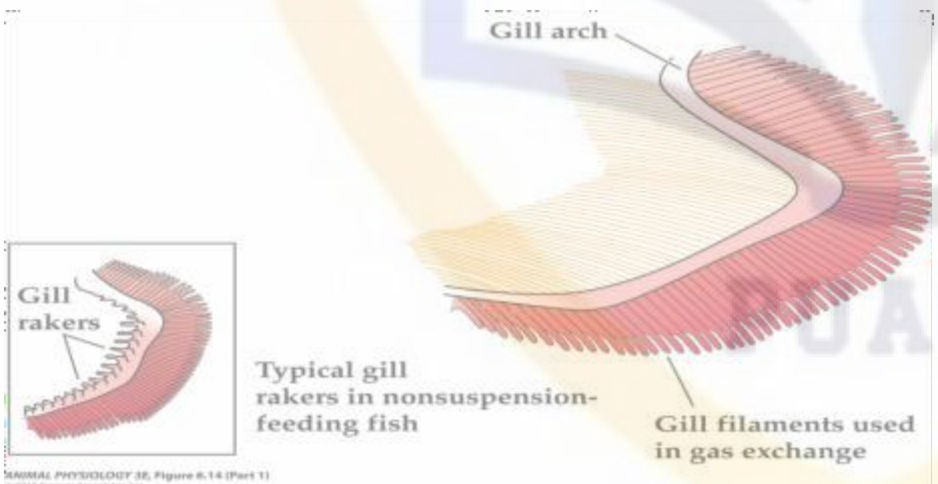
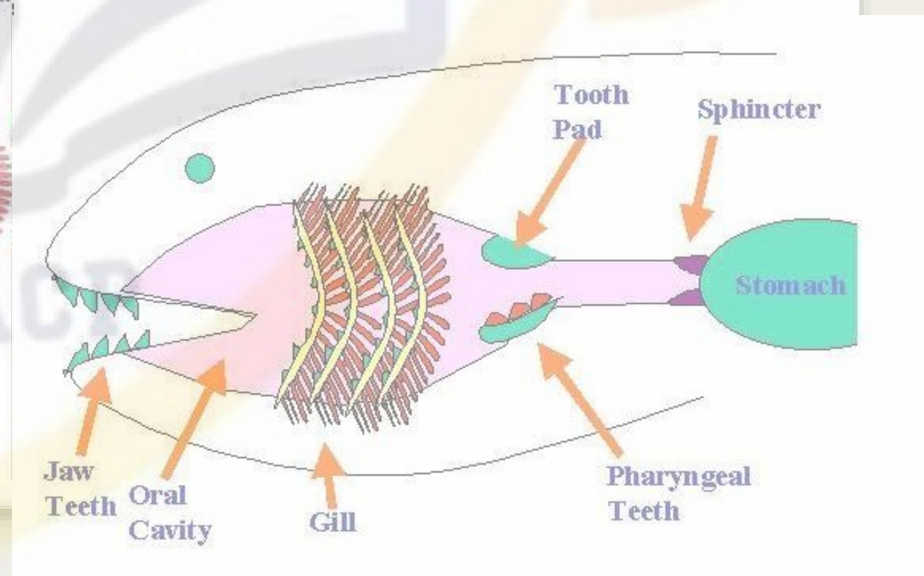
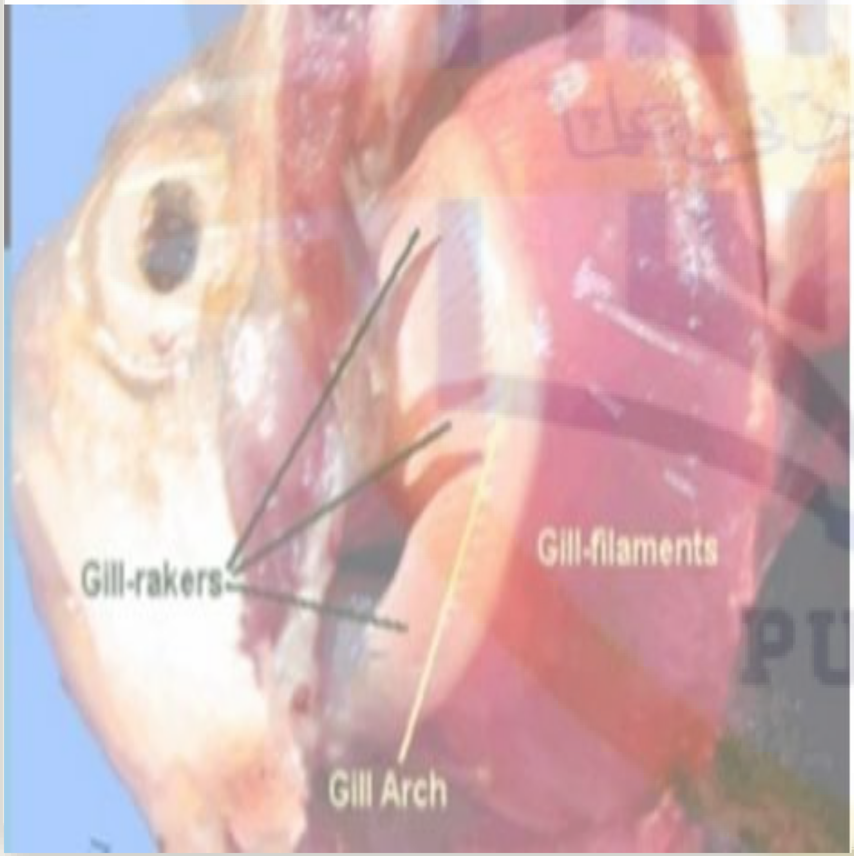
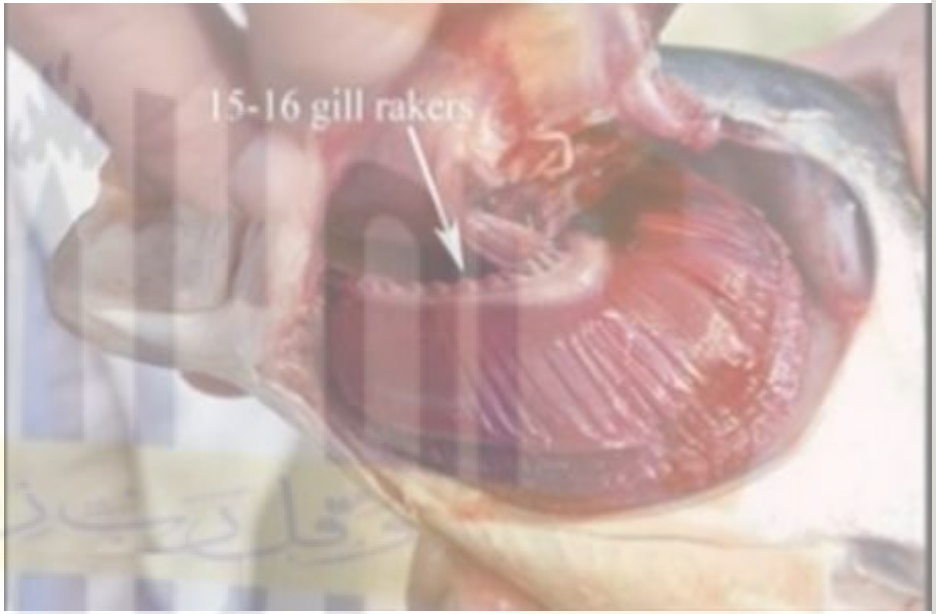




OMNIVOROUS

In the omnivorous fishes like *Puntius sarana,*gill rakers are short and stumpy.

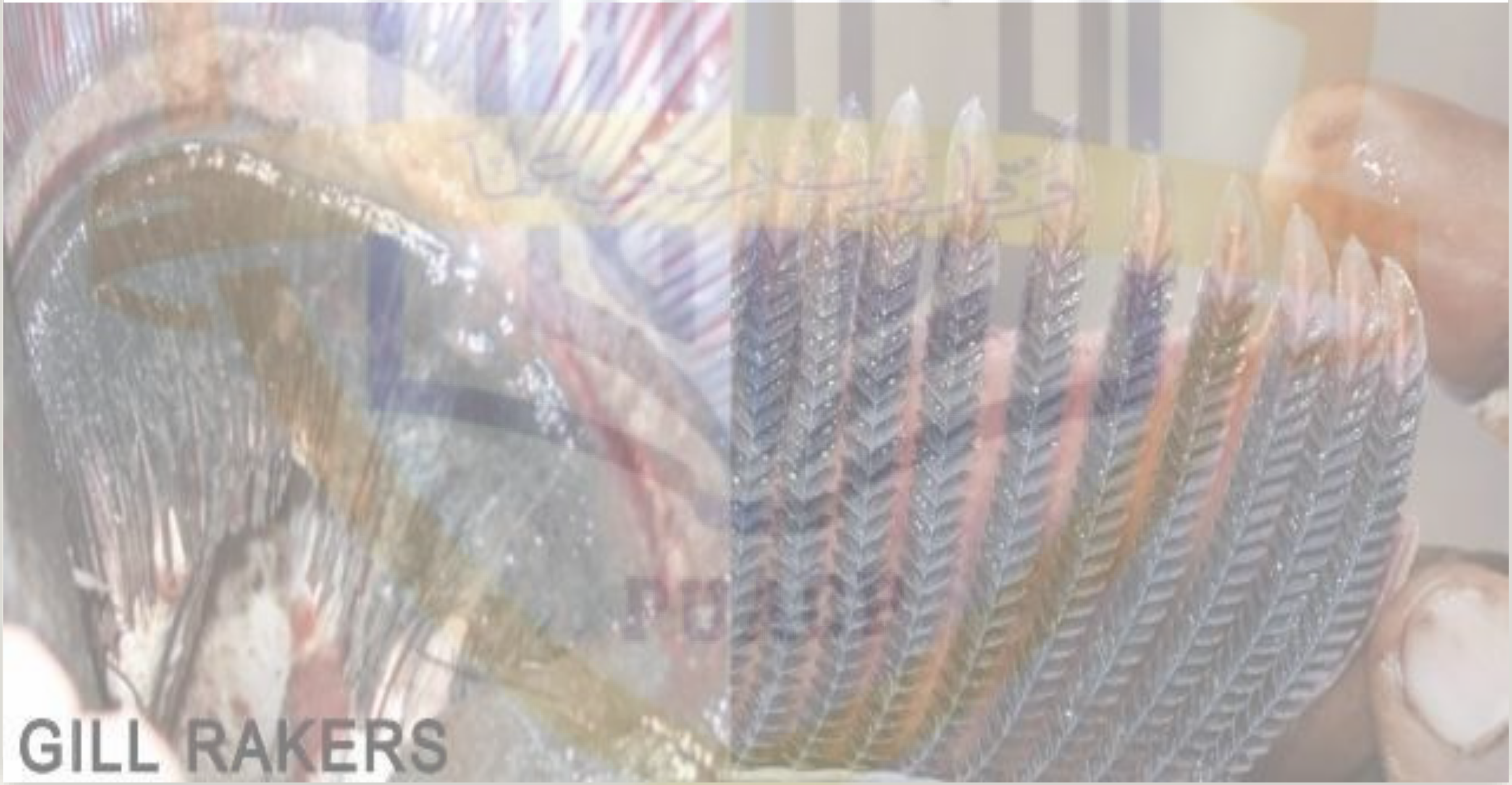




HERBIVOROUS:

Like *Labeo rohita,Cirrhina marigala,*gill rakers form a broad sieve like structure across the gill slits for filtering the water in order to retain the food in the bucco-pharynx

Carnivorous Fishes : Gill rakers are normally long, hard and teeth like forming rasping organs as in *Wallago attu, Mystus seenghala, Channa striatus.*



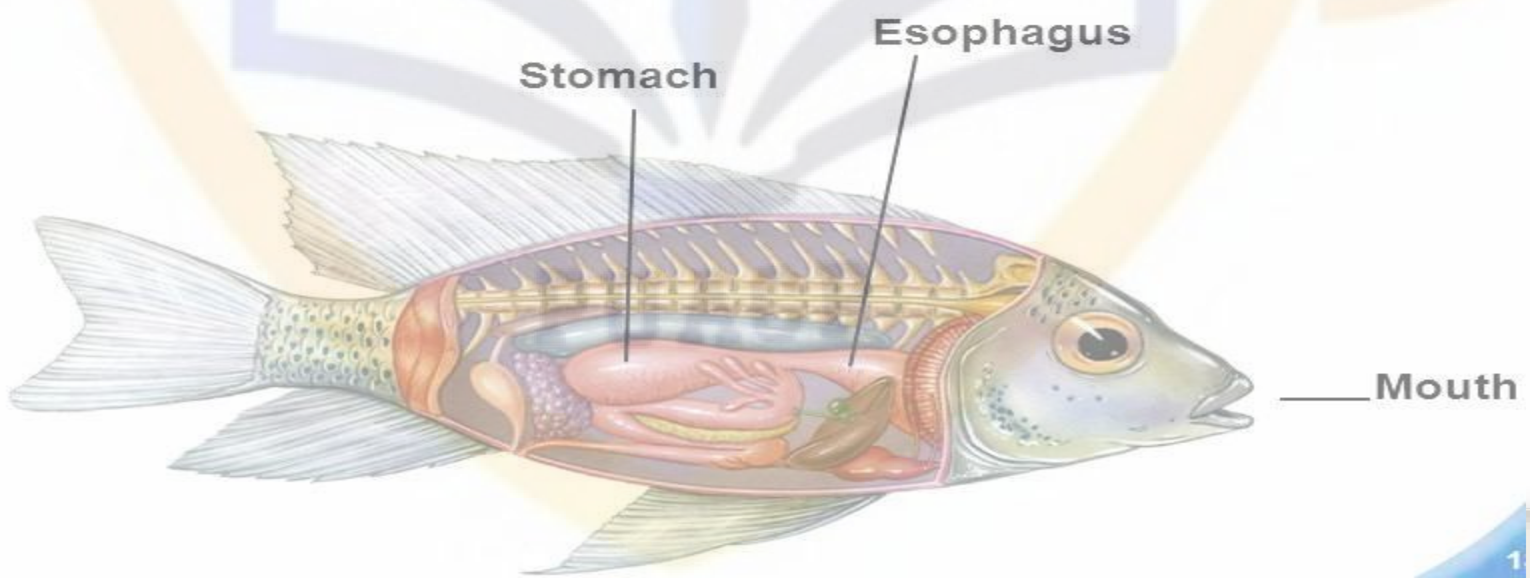
❖Oesophagus

•The pharynx opens behind into the oesophagus which have large number of mucus secreting cells which are scattered in the mucosa.

•Taste buds are also present e.g. *Labeo rohita, C.carpio, Catla catla*

• Short & narrow tube in case of herbivorous and omnivorous fishes e.g. *Labeo rohita,Puntius sarana, Cyprinus carpio etc*

•Large & distensible tube in case of carnivores and predatory fishes (e.g. *Wallago attu, H. fossilis* etc



❖Stomach

- Stomach acquire different shapes according to the availability of space in the body cavities of different fishes.

STOMACH → ANTERIOR – CARDIAC STOMACH

STOMACH → POSTERIOR – PYLORIC STOMACH

- All the fishes do not possess a true stomach as it is almost absent in various herbivorous fishes like *Labeo rohita,Catla catla* etc.

- In such fishes, the anterior part of the intestine is swollen to form a sac like structure called intestinal bulb.

- A true stomach is present in the carnivorous and predatory fishes e.g. *Channa striatus, Mystus seenghala* etc .

Carnivorous fishes:

- Stomach is generally sac-like and thick walled in Carnivorous and predatory fishes.

Omnivorous fishes:

•Stomach of omnivorous fishes is also sac like .e.g. *Puntius sophore,Cyprinus carpio* etc.
•In some fishes like *Hilsa hilsa,Gudusia chapra,* stomach is reduced in size but is greatly thickened to become gizzard like for trituration of food

STOMACH OF CARNIVOROUS & OMNIVOROUS FISHES

❖Pyloric caeca

- Anterior part of the intestine give rise to a number of finger-like outgrowths called pyloric or intestinal caeca.

- Pyloric caeca serve as accessory food reservoirs .

- Histologically, intestinal caeca resembles the intestine and probably serve to enhance the absorptive area.

❖Intestine

•The part of alimentary canal that follows the stomach is called intestine and is divided into two parts:

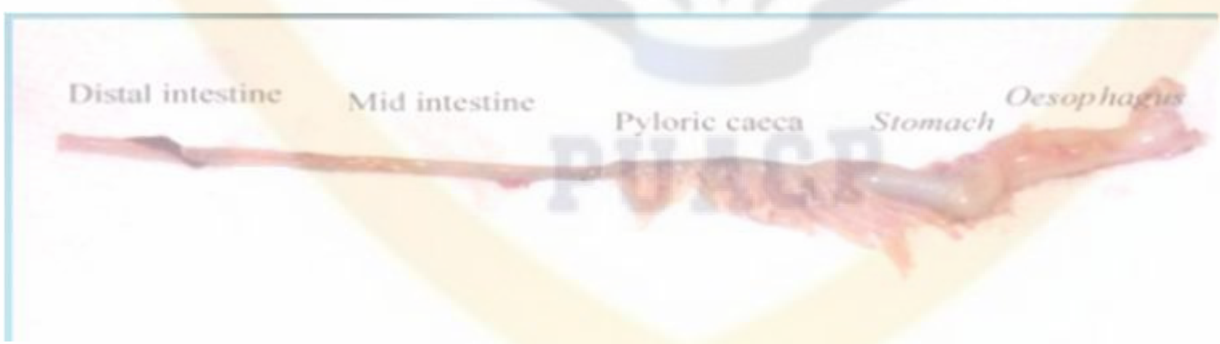
1. Anterior part : small intestine
2. Posterior part : large intestine

•The small intestine just behind the stomach receives ducts from the liver and pancreas is called as duodenum while rest part is called ileum .

•There is no clear cut demarcation between the small intestine and large intestine .

•The length of the intestine depends upon the feeding habit of the fish.

Carnivorous fishes : It is shortened in carnivores such as in *Wallogo attu, Mystus seenghala* because flesh can be digested more readily than the plant based food stuff.



Herbivorous fishes:

•Intestine is often elongated and arranged in many folds in case of herbivorous fishes.

• Longer intestines are of great advantage to herbivorous fishes as they retain food for long period of time to ensure digestion.

STRUCTURES OF THE DIGESTIVE TRACT OF HERBIOUROUS FISH: (A) ESOPHAGUS, (B) STOMACH, (B1) CARDIAC STOMACH, (B2) FUNDIC STOMACH, (B3) PYLORIC STOMACH, (C) PYLORIC CAECA, (D) ANTERIOR INTESTINE, (E) MIDINTESTINE, AND (F) POSTERIOR INTESTINE.

Omnivorous fishes:

- Intermediate length is found in case of omnivorous fishes e.g. *Puntius sophore* etc.

- The intestinal bulb of Rohu is about 25cm,the small intestine about 8m and the large intestine about a meter in length.

❖Rectum

•It is not usually distinguishable externally but an ileo-recta valve is present in few species of fish to demarcate it from the intestine e.g. *Tetradon*

•Histologically ,the mucosal folds of the rectum differ from the intestine in being shorter and broader, possess a large number of mucus secreting cells produce copious mucus to lubricate waste food and aid in easy defecation

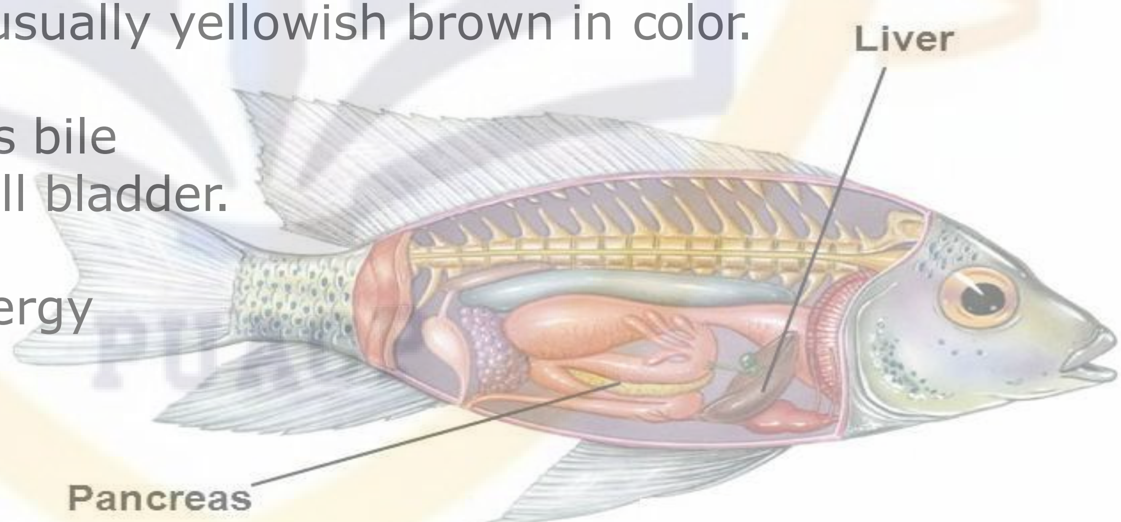
❖Digestive glands

Two main digestive glands :

1 . Pancreas: pancreas is a diffuse gland, but is well developed around the blood vessels between the lobes of the liver.

Pancreas has two digestive functions:
1. Source of exocrine secretion into the intestine.
2. Endocrine secretion of the hormones insulin and glucagon .

2.Liver:
- Liver is a bilobed gland usually yellowish brown in color.
- The liver in fish produces bile which is stored in the gall bladder.
- Key storage of food energy in the form of glycogen.



❖ Digestive enzymes

- Pepsin
- Trypsin
- Chymotrypsin
- Pancreatic lipase
- Ribonucleases
- Deoxyribonucleases

Fish Digestive Physiology

Digestion is accomplished in

Stomach

- Low pH - HCl, other acids (2.0 for some tilapia!)
- Proteolytic enzymes (mostly pepsin)

Digestive Processes: Stomach

- Catfish as an example - its digestive processes are similar to that of most monogastric animals
- Food enters stomach, neural and hormonal processes stimulate digestive secretions
- As stomach distends, parietal cells in lining secrete **gastrin**, assisting in digestion
- Gastrin converts the zymogen **pepsinogen** to **pepsin** (a major proteolytic enzyme)
- Some fish have **cirulein** instead of gastrin

Digestive Processes: Stomach

- Flow of **digesta** out of stomach is controlled by the pyloric sphincter
- **Pepsin** has pH optimum and lyses protein into small peptides for easier absorption
- Minerals are solubilized; however, no lipid or COH is modified
- Mixture of gastric juices, digesta, mucous is known as **chyme**

Fish Digestive Physiology

Digestion is accomplished in

Stomach

Intestine

- Alkaline pH (7.0 - 9.0)
- Proteolytic enzymes - from pancreas & intestine
- Amylases (carbohydrate digestion) - from pancreas & intestine
- Lipases (lipid digestion) - from pancreas & liver (gall bladder, bile duct)

Digestive Processes: Intestine

1. **Chyme** entering the small intestine stimulates secretions from the pancreas and gall bladder (**bile**)
2. Bile contains salts, cholesterol, phospholipids, pigments, etc.
3. Pancreatic secretions include bicarbonates which buffer acidity of the chyme
4. Zymogens for proteins, COH, lipids, chitin and nucleotides are secreted
5. e.g., **enterokinase** (trypsinogen --> trypsin)
6. Others: chymotrypsin, carboxypeptidase, aminopeptidase, chitinase

Digestive Processes: Intestine

- Digestion of carbohydrates is via **amylase**, which hydrolyzes starch
- Others: **nuclease, lipase**
- **Cellulase**: interesting in that it is not secreted by pancreas, but rather produced by gut bacteria
- Note: intestinal mucosa also secretes digestive enzymes

Fish digestive physiology

Absorption is accomplished in

Intestine

- Diffusion into mucosal cells
- Phagocytosis/pinocytosis by mucosal cells
- Active transport via carrier molecules

Mooneye- *Hiodon Tergisus*
averages 11 - 15 inches

Digestive processes: Absorption

- Most nutrient absorption occurs in the intestine
- Cross-section of the intestinal luma shows that it is highly convoluted, increasing surface area
- Absorption through membrane is either by **passive diffusion** (concentration gradient)
- Or by **active transport** (requires ATP)
- Or via **pinocytosis** (particle engulfed)
- Nutrients absorbed by passive diffusion include: electrolytes, monosaccharides, some vitamins, smaller amino acids

Digestive processes: absorption

- Proteins are absorbed primarily as amino acids, **dipeptides** or **tripeptides**
- Triglycerides are absorbed as **micelles**
- COH's absorbed as **monosaccharides** (e.g., glucose, except for crustaceans)
- Calcium and phosphorus are usually complexed together for absorption
- All nutrients, excluding some lipids, are absorbed from the intestine via the hepatic portal vein to the liver


THANK YOU !